

Literature Review for Smart Ageing Research

From a Broad Topic to a Defensible Research Question

Week	Content Coverage	Date	
1	Introduction to Research Methodology and Ethics	2026/9/5	2026/9/5
2	Literature Review	2026/9/12	2026/9/18下午
3	Learning LaTeX & Writing Research Proposals I	2026/9/19	2026/9/19
4	Writing Research Proposals II	2026/9/26	2026/10/10
5	Writing Research Papers I	2026/10/3	2026/10/10晚上
6	Writing Research Papers II	2026/10/10	2026/10/17
7	Writing Research Papers III	2026/10/17	2026/10/17晚上
8	Research Ethics	2026/10/24	2026/10/24
9	Making Posters and Slides	2026/10/31	2026/10/31
10	Making Posters and Slides	2026/11/7	2026/11/7
11	Making Posters and Slides	2026/11/14	2026/11/14
12	Individual Student Presentation I	2026/11/21	2026/11/21
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14	Individual student presentation III	2026/12/5	2026/12/5
15	Individual student presentation IV	2026/12/12	2026/12/12

Course Outline

Module	What this module establishes
01 Introduction	Diagnose weak review writing and create an initial synthesis claim
02 Purpose and Review Types	Match the review approach to the research decision
03 Question and Scope	Define an answerable question, boundaries and protocol
04 Search Strategy	Build a reproducible search across relevant disciplines
05 Screening and Extraction	Apply eligibility rules and capture comparable evidence
06 Critical Appraisal	Judge credibility with design-specific criteria
07 Synthesis and Gap	Explain evidence patterns and derive a bounded research gap
08 Take-home Messages	Integrate the decisions in a one-page mini-review protocol

Each stage sets the boundaries for the decisions that follow

01

Introduction

What makes a literature review claim defensible?

Activity 1: What Is Wrong With This Review Paragraph?

Deliverable: one revised sentence.

Smith et al. (2021) used wearable accelerometers and achieved 89% accuracy. Jones et al. (2022) used smart-home sensors and achieved an AUROC of 0.84. Lee et al. (2024) combined both modalities and reported the best performance. These studies show that AI is promising for older adults.

In pairs

- Underline one claim that the paragraph cannot yet defend
- Identify one missing comparison and one missing boundary
- Rewrite the first sentence as a synthesis claim

Use the paragraph as teaching material. The study details are illustrative.

Sample Answer: A High Metric Does Not Establish a Conclusion

Problems

- Different outcomes and metrics may not be comparable
- Cohorts and validation designs remain unspecified
- The claim of best performance assumes a common benchmark
- Accuracy appears without calibration or uncertainty
- Promising has no connection to a care decision

Possible rewrite

Reported discrimination was moderate to high, but direct comparison was not justified because the target outcomes, cohorts and validation designs differed.

Synthesis begins by establishing what can legitimately be compared

Sample Answer: A High Metric Does Not Establish a Conclusion

Possible rewrite

Reported discrimination was moderate to high, but direct comparison was not justified because the target outcomes, cohorts and validation designs differed. Smith et al. (2022) evaluated wearable accelerometers, whereas Jones et al. (2022) examined smart-home sensors using a different performance metric. Although Lee et al. (2022) reported stronger numerical performance for a multimodal approach, the absence of a common dataset and validation protocol prevents a conclusion that this method is superior. Collectively, these studies support the feasibility of sensor-based monitoring, but its comparative and clinical value remains uncertain.

Synthesis claims are the building blocks of the literature review's argument.

02

Purpose and Review Types

What decision should this review support?

The Research Decision



Possible doctoral decisions

- Is a new study justified?
- Which population or outcome is underrepresented?
- Which method is credible enough to build on?
- What needs validation before implementation?

The value of a review lies in the decision it improves

Catalogue and Critical Synthesis

Catalogue	Critical synthesis
Reports one paper at a time	Organizes evidence around a question
Repeats authors' conclusions	Evaluates the basis of each claim
Lists different results	Explains agreement and disagreement
Ends with a general call for more research	Specifies the missing evidence and its consequence

Compare Explain Bound Conclude

A strong review changes the unit of writing from paper to claim

Running Case: AI-Enabled Home Monitoring

Broad topic

AI-enabled home monitoring for healthy ageing

Research decision

Is a prospective multisite validation study justified, and what should it measure?

Initial boundaries

- Community-dwelling adults aged 65 years or older
- Wearable or ambient passive sensing
- Six-month functional decline or falls
- Model validity, usability and equity

All study examples and numerical results in the activities are illustrative.

One running case connects every stage of the review

Review Types | 常见综述类型

Review type	核心问题与适用情境
Scoping review **范围综述**	What evidence exists? 适合范围宽、概念或方法尚未稳定的领域
Systematic review **系统综述**	What does the evidence show for a focused question? 适合明确的效果、诊断或预测问题
Systematic map **系统证据地图**	Where is evidence concentrated or absent? 适合描绘大型、异质证据领域
Qualitative evidence synthesis **定性证据综合**	How do people experience or understand an issue? 适合体验、意义与接受度
Realist review **现实主义综述**	What works, for whom, in what context, and why? 适合复杂干预与实施机制
Critical narrative review **批判性叙述综述**	How should concepts or assumptions be interpreted? 适合理论、概念和批判性论证

Meta-analysis 元分析 is a statistical synthesis method, not a review type.

Choose the review type from the question and intended evidence product

JBI Manual for Evidence Synthesis (2024). Cochrane Handbook v6.5 (2024).

Review Type and Decision Need

Decision need	Likely review approach
Map concepts, technologies or evidence volume	Scoping review
Estimate intervention effects	Systematic review
Evaluate prediction model evidence	Focused systematic review or systematic map
Understand experiences and meanings	Qualitative evidence synthesis
Explain how context changes implementation	Realist review
Reframe a field conceptually	Critical narrative review

Meta-analysis is a synthesis method rather than a review type.

The question determines the review type

JBI Manual for Evidence Synthesis (2024). Cochrane Handbook v6.5 (2024).

Activity 2: Which Review Type Fits?

Deliverable: review type and decision rationale.

Case	Question
A	What monitoring technologies and outcomes for older adults have been studied, and for which purposes?
B	Does remote monitoring reduce unplanned hospital admissions?
C	How do family caregivers experience passive home monitoring?
D	Why does the same monitoring programme succeed in some care systems but fail in others?

Choose the best starting approach and justify the choice in one sentence.

Sample Answer: | 同一智慧康养主题的四种研究任务

Decision 决策	Best starting review	What the team actually does 实际操作
Which monitoring technologies and outcomes have been studied? 研究过哪些监测技术和结局?	Scoping review 范围综述	Search broadly, classify technologies and outcomes, identify empty areas 广泛检索，分类技术与结局，识别空白区域
Does remote monitoring reduce unplanned admissions? 远程监测是否减少非计划住院?	Systematic review 系统综述	Fix eligibility and outcomes, assess bias, estimate effects if comparable 固定纳排与结局，评价偏倚，可比时估计效果
How do family caregivers experience passive monitoring? 家庭照护者如何体验被动监测?	Qualitative evidence synthesis 定性证据综合	Include qualitative studies, code experiences, interpret themes 纳入定性研究，编码体验，解释主题
Why does implementation work in some care systems? 为何在某些照护系统中实施成功?	Realist review 现实主义综述	Develop and test context-mechanism-outcome explanations 建立并检验情境、机制与结果解释

The topic stays the same; the decision changes the evidence workflow

03

Question and Scope

What exactly must the evidence be able to answer?

From Topic to Answerable Question

Topic
Technology and ageing

Problem
Late detection of decline

Decision
Whether and how to validate

Review question
Evidence and limits

Decision test

Would the answer change the study design or a care decision?

Example review question

What evidence supports sensor-based prediction, and what limits its care utility?

A useful question defines the answer that would change a decision

PICO and PCC | 两种常用问题框架

Framework	Elements	Best suited to	Running-case example
PICO	P Population 人群 I Intervention 干预 C Comparator 对照 O Outcome 结局	干预效果问题 Does an intervention work?	Older adults + remote monitoring + usual care + falls
PCC	P Population 人群 C Concept 核心概念 C Context 情境	范围综述问题 What evidence exists?	Older adults + passive monitoring + home or community care

PICO asks whether an intervention changes an outcome.
PCC maps a population, concept and context.

Use a framework to expose missing elements, not to force every question into PICO

JBI Manual for Evidence Synthesis (2024).

Question Frameworks

Question family	Useful structure
Intervention effect (PICO)	Population, Intervention, Comparator, Outcome
Evidence landscape (PCC)	Population, Concept, Context
Prediction	Population, index model, outcome, horizon, setting
Experience	Population, phenomenon, context
Implementation	Context, mechanism, outcome

A framework exposes missing elements. It should not force every question into PICO.

The question family determines which elements need definition

JBI Manual for Evidence Synthesis (2024).

Scope Boundaries and Units of Evidence

Define before screening

- Population and setting
- Technology, intervention or phenomenon
- Target outcome and time horizon
- Eligible study designs
- Publication status, dates and languages
- Unit of evidence: paper, cohort, dataset or model

Smart-ageing caution

One paper may report many models. Several papers may reuse the same cohort or dataset.

Duplicated cohorts can create a false impression of independent evidence.

Scope boundaries function as analytical assumptions

Activity 3: Narrow the Running Case

Deliverable: one question and one exclusion criterion.

Starting topic

AI monitoring for healthy ageing

Step	Boundary
1	Decision
2	Population
3	Technology or phenomenon
4	Target outcome
5	Time horizon
6	Setting
7	Eligible evidence

The exclusion criterion must support consistent decisions by two reviewers.

Sample Answer: Operational Boundaries Make a Question Reviewable

Possible question

Among community-dwelling adults aged 65 years or older, what evidence supports wearable or ambient sensor-based models for predicting functional decline or falls within six months, and how do validation design, usability and equity affect their potential care utility?

Example exclusion

Laboratory-only fall-detection studies involving younger adults without age-stratified results.

The question, eligibility criteria and synthesis must describe the same study universe

The Minimum Viable Protocol

Specify before searching

- Rationale, decision and review type
- Review question and eligibility criteria
- Information sources and draft search
- Selection and conflict resolution
- Extraction and appraisal
- Planned synthesis and handling of deviations

Registration and change control

Use an appropriate registry or open repository when applicable. Record every protocol change with its date and rationale.

A protocol reduces hindsight-driven decisions.

A transparent protocol preserves the logic of the review

JBI Manual for Evidence Synthesis (2024). PRISMA-P (2015).

04

Search Strategy

Where would relevant evidence be indexed, and how would each field describe it?

The Evidence Ecosystem

Health and care

- MEDLINE or PubMed
- Embase
- CINAHL
- PsycINFO

Engineering and computing

- IEEE Xplore
- ACM Digital Library
- Relevant conferences
- Technical standards

Cross-disciplinary sources

- Scopus or Web of Science
- Trial registries
- Theses and grey literature
- Citation searching

Database coverage should mirror the disciplines in the question

Databases and Search Terms | 数据库与检索缩写

Term	Full name or meaning	Why it appears in this course
MEDLINE / PubMed	MEDLINE 医学文献数据库；PubMed 是常用检索平台	Health, medicine and ageing research
CINAHL	Cumulative Index to Nursing and Allied Health Literature	Nursing, care and allied health
PsycINFO	APA psychology and behavioural-science database	Psychology, behaviour and acceptability
IEEE Xplore	Institute of Electrical and Electronics Engineers platform	Engineering, sensors and computing
ACM Digital Library	Association for Computing Machinery Digital Library	Computer science and human-computer interaction
MeSH	Medical Subject Headings 医学主题词	Controlled vocabulary used in MEDLINE
PRESS	Peer Review of Electronic Search Strategies	Peer review of a core search strategy
PRISMA-S	PRISMA extension for reporting literature searches	Transparent reporting of search methods

Choose each information source because it covers a necessary part of the question

PRESS 2015. PRISMA-S (2021).

Concept Blocks and Vocabulary

Block	Illustrative vocabulary	Role
Population	older adults, aged, ageing, aging, geriatric	Core
Monitoring technology	smart home, wearable, ambient sensor, passive sensing	Core
Target	functional decline, activities of daily living, falls	Core
Analytical approach	machine learning, artificial intelligence, prediction model	Optional

Combine synonyms with OR. Connect different concepts with AND.

Search sensitivity depends on vocabulary coverage

Database-Specific Search Syntax

Illustrative PubMed-style query

```
("Aged"[MeSH] OR older adult*[tiab]
 OR ageing[tiab] OR aging[tiab])
AND
("Monitoring, Ambulatory"[MeSH]
 OR "smart home"[tiab] OR wearab*[tiab]
 OR "ambient sensor*"[tiab])
AND
("Activities of Daily Living"[MeSH]
 OR "functional decline"[tiab] OR fall*[tiab])
```

Validation questions

- Does the query retrieve known relevant papers?
- Which terms appear in titles, abstracts and indexing?
- Which irrelevant records dominate the results?
- How must the syntax change for the next database?

Every database requires translation, testing and revision

Cochrane Handbook v6.5, Chapter 4 (2024). PRISMA-S (2021).

Supplementary Search and Documentation

Supplementary discovery

- Backward and forward citation searching
- Hand-searching key journals and conferences
- Registries, theses and relevant grey literature
- Preprints and conference papers as distinct evidence classes
- Author contact for critical missing information

Minimum search log

Field	Example
Source and platform	MEDLINE via PubMed
Date	16 Sep 2026
Query	Full version saved
Limits	None in the first run
Results	Count before deduplication
Revision	Reason for each change

A reproducible search records the query and every route used to find records

PRESS 2015. PRISMA-S (2021).

Activity 4: How Do You Search for Papers?

Deliverable: one group workflow and one improvement.

Discuss	Share your actual practice
1	Where do you usually start: a database, Google Scholar, reference list or AI tool?
2	How do you choose keywords and discover alternative terms?
3	How do you decide whether a paper is relevant?
4	When do you stop searching, and do you follow citations or related papers?
5	What do you save so that you or another researcher can repeat the search?

Compare your habits. Identify one strength, one blind spot and one practical improvement.

Debrief: From Search Habits to a Reproducible Process

Common habit	What it helps	Reproducible upgrade
Start with one familiar platform	Fast orientation	Match databases to the evidence ecosystem
Use a few intuitive keywords	Find obvious papers	Build concept blocks with synonyms and subject headings
Follow citations from a key paper	Discover related terminology	Record backward and forward citation routes
Stop when results feel repetitive	Keeps the task manageable	Define coverage checks and the last-search date
Save PDFs or citations only	Supports reading	Save platform, full query, date, limits and result count

The interface may build the syntax, but the researcher must understand, test and save the search.

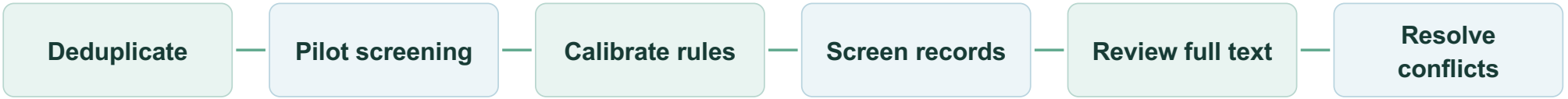
Keep the speed of familiar habits, then add coverage checks and a reproducible record

05

Screening and Extraction

Which records belong, and what information must be captured for comparison?

Operational Eligibility Rules



Vague rule

Include studies of older people

Operational rule

Include studies with mean age of at least 65 years, or separately reported results for participants aged 65 years or older.

Record reviewer decisions, exclusion reasons, conflict resolution and PRISMA flow counts.

Eligibility criteria should produce consistent screening decisions

Page et al. PRISMA 2020. BMJ. 2021;372:n71.

Data Extraction for Synthesis

Domain	Example fields	Why it matters
Context	Country, setting, care pathway	Defines transferability
Population	Age, frailty, cognition, digital access	Defines representation and equity
Data	Sensor, observation window, missingness	Defines measurement conditions
Model	Target, predictors, algorithm	Clarifies the estimand and method
Evaluation	Split unit, validation, metrics, uncertainty	Determines credibility
Utility	Calibration, threshold, usability, workflow	Connects performance to use

Separate author-reported information from reviewer judgment.

Extract the information needed to answer, compare and appraise

Activity 5: Include, Exclude or Uncertain?

Apply the running-case criteria.

Record	Abstract information	Decision	Reason code
A	84 adults aged 70–91, wearable data, six-month falls, participant-level CV		
B	Laboratory fall detection in healthy volunteers aged 18–35		
C	Home-care cohort aged 45–92, predicts ADL decline, no age-stratified abstract result		
D	Interviews with 25 caregivers about acceptability of passive sensors		

For uncertain records, state what additional information you need.

Sample Answer: Uncertain Is a Valid Screening State

Record	Decision	Rationale
A	Include	Eligible population, technology and prospective target. Flag internal validation.
B	Exclude	Wrong population and laboratory-only context.
C	Uncertain	Retrieve the full text or age-stratified information.
D	Exclude from prediction review	Tag as linked implementation evidence.

Screening applies the scope rules. Appraisal evaluates credibility after inclusion.

An uncertain decision protects sensitivity when the abstract is incomplete

06

Critical Appraisal

How credible and applicable is the evidence after inclusion?

Appraisal Tools by Study Claim

Study claim	Appraisal family
Randomized intervention effect	RoB 2
Non-randomized intervention effect	ROBINS-I
Diagnostic accuracy	QUADAS-3
Prediction model development or evaluation	PROBAST+AI
Qualitative experience	JBI or CASP qualitative tools

TRIPOD+AI supports transparent reporting.
PROBAST+AI assesses quality, risk of bias and applicability.

The study design and claim determine the appraisal tool

TRIPOD+AI (BMJ 2024). PROBAST+AI (BMJ 2025). QUADAS-3 (Ann Intern Med 2026).

Reporting and Appraisal Tools | 报告规范与评价工具

Tool	Full name	Primary use
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses	Reporting a completed systematic review
PRISMA-P	PRISMA extension for Protocols	Reporting a review protocol
RoB 2	Cochrane Risk of Bias tool, version 2	Randomized trials
ROBINS-I	Risk Of Bias In Non-randomized Studies of Interventions	Non-randomized intervention studies
QUADAS-3	Quality Assessment of Diagnostic Accuracy Studies, version 3	Diagnostic-accuracy studies
TRIPOD+AI	Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis + AI	Reporting prediction-model studies
PROBAST+AI	Prediction model Risk Of Bias ASsessment Tool + AI	Risk of bias and applicability of prediction models

A reporting guideline does not replace a risk-of-bias assessment

PRISMA 2020. TRIPOD+AI (2024). PROBAST+AI (2025). QUADAS-3 (2026).

Validity Threats in Smart-Ageing AI Studies

Data and analytical pipeline

- Participant or household leakage across data splits
- Preprocessing or feature selection outside training folds
- Repeated tuning on the test set
- Inadequate handling of missing sensor data

Clinical evaluation and deployment

- Unclear outcome or reference standard
- Discrimination without calibration
- No temporal, geographic or site validation
- Missing subgroup, usability or workflow evaluation

An easy validation design can inflate apparent performance.

Predictive performance alone does not establish clinical value, transportability or fairness

Collins et al. TRIPOD+AI. BMJ. 2024;385:e078378. Moons et al. PROBAST+AI. BMJ. 2025;388:e082505.

Metrics and Clinical Terms | 指标与临床术语

Term	Meaning	Interpretation in this course
ADL	Activities of Daily Living 日常生活活动	A functional outcome, not a single universal measure
AUROC	Area Under the Receiver Operating Characteristic Curve	Discrimination: ranking higher-risk and lower-risk cases
CV	Cross-validation 交叉验证	Internal validation; independence depends on the split unit
Discrimination	区分高风险与低风险个体的能力	Does not show whether predicted probabilities are accurate
Calibration	预测概率与实际发生率的一致程度	Essential when probabilities guide decisions
External validation	在独立地点、时间或人群中检验模型	Tests transportability beyond the development data
Clinical utility	模型是否改善实际决策或结局	Requires more than a high performance metric

Prediction accuracy, calibration and clinical utility answer different questions

Activity 6: Which Study Provides Stronger Decision Evidence?

Use at least three appraisal questions.

Study A

5,000 participants
AUROC 0.92
Random window-level split
Same participants in training and test data
No calibration or external validation

Study B

650 participants
AUROC 0.79
Site-held-out validation
Calibration and prespecified threshold
Frailty subgroup analysis
Prospective silent deployment

Which study better supports a care decision, and why?

Sample Answer: Credibility Extends Beyond Sample Size and AUROC

Why Study B provides stronger decision evidence

- The validation site is independent
- Calibration and a decision threshold are reported
- Frailty-related performance is examined
- Prospective silent deployment tests workflow realism

Study A

The window-level split allows participant leakage and can inflate apparent performance.

Study B still requires outcome and workflow evaluation before deployment.

Performance reflects the model, data, validation design and deployment context

Critical Appraisal Changes by Review Type | 质量评价因类型而变

Review type	Role of appraisal 评价的作用	Typical focus 常见关注点
Scoping review / map	Optional or limited, depending on purpose 可选或有限，取决于目的	Evidence characteristics, coverage and limitations 证据特征、覆盖范围与局限
Focused systematic review	Usually central to the conclusion 通常直接影响结论	Risk of bias, applicability and certainty 偏倚风险、适用性与证据确定性
Qualitative synthesis	Judges methodological limitations and contribution 判断方法局限及其对综合的贡献	Sampling, data collection, reflexivity and interpretation 抽样、资料收集、反思性与解释
Realist / critical narrative	Tests relevance and rigour for the argument 检验材料对论证的相关性与严谨性	Whether evidence can support or challenge the proposed explanation 证据能否支持或挑战解释

The appraisal question must match the claim that the review intends to make

Tools depend on study design and claim, for example RoB 2, ROBINS-I, PROBAST+AI and CASP.

07

Synthesis and Gap

What pattern remains after credibility and context are considered?

Claim-Centred Synthesis

Return to Activity 1: test the opening synthesis claim against evidence, credibility, explanation and boundary.

Review dimension	Evidence pattern	Credibility	Why findings differ	Implication
Validation				
Calibration				
Equity				
Workflow utility				

Claim — **Evidence** — **Explanation** — **Boundary**

Group studies by the question they inform

Campbell et al. SWiM guideline. BMJ. 2020;368:l6890. Cochrane Handbook v6.5 (2024).

Heterogeneity and the Research Gap

Clinical
Frailty, cognition, disease burden

Technical
Sensor, sampling, feature construction

Methodological
Outcome definition, split and metric

Contextual
Home environment, country and care pathway

A defensible gap connects four elements

Known pattern — **Residual uncertainty** — **Consequence** — **Feasible next design**

Study scarcity becomes a gap when it leaves a consequential question unresolved

Activity 7: Rebuild the Opening Synthesis Claim

Return to Activity 1. Write individually, revise with a peer, then hear two read-outs.

Study	Design	Result	Key limitation
A	n=220, internal participant-level CV	AUROC 0.86	No calibration
B	n=680, temporal validation	AUROC 0.76, calibration slope 0.95	Lower sensitivity in frail adults
C	n=310, external-site validation	AUROC 0.74	28% sensor missingness, no workflow test

Rebuild it in four sentences

- Convergent finding
- Apparent contradiction
- Plausible explanation
- Bounded gap and next research question

Illustrative teaching data.

Sample Answer: A Gap Should Lead to a Better Study

Across the three studies, passive sensing showed moderate discrimination for six-month decline, suggesting a reproducible signal without deployment-ready evidence. The highest AUROC arose from internal validation, whereas temporal and external-site estimates were lower but more transportable. Validation stringency, case mix and sensor missingness may explain the difference, while only one study reported calibration. The defensible gap concerns prospective multisite evaluation of calibration, performance in frail adults, missing-data robustness and care-workflow consequences.

Possible next question

Can a prespecified sensor model remain calibrated and clinically useful across community sites with different frailty profiles and sensor-completion rates?

A strong gap identifies the missing decision-relevant test

Synthesis and Final Evidence Product | 综合方法与最终产出

Review type	Synthesis 综合	Final product 最终证据产品
Scoping review	Descriptive counts plus thematic or conceptual mapping 描述性统计及主题或概念归类	Map of concepts, methods and gaps 概念、方法与空白地图
Systematic review	Narrative synthesis; meta-analysis when studies are sufficiently comparable 叙述综合；可比性足够时进行元分析	Bounded conclusion on effect, accuracy or prognosis 有边界的效果、准确性或预后结论
Systematic map	Structured coding and evidence matrix 结构化编码与证据矩阵	Visual or tabular distribution of evidence 证据分布图或表
Qualitative synthesis	Themes, concepts or explanatory interpretation 主题、概念或解释性综合	Understanding of experiences and meanings 对体验与意义的理解
Realist review	Context-mechanism-outcome explanations 情境、机制与结果解释	Refined programme theory 经修正的项目理论
Critical narrative review	Critical comparison of concepts and assumptions 批判比较概念与假设	Conceptual argument or reframing 概念论证或重新界定

Meta-analysis is one possible synthesis method, not a review type or a compulsory endpoint

08

Take-home Messages

How will the complete review logic be documented and carried into the next study?

Review-Type Decision Check | 综述类型决策检查

1	**Decision**	What decision will the review support? 综述要支持什么决策?
2	**Evidence unit**	Effects, study characteristics, experiences, mechanisms or concepts? 要综合效应、研究特征、体验、机制还是概念?
3	**Search logic**	Broad mapping, exhaustive focused search or iterative theory testing? 广泛描绘、聚焦穷尽，还是理论驱动的迭代检索?
4	**Credibility**	Which appraisal criteria can test the intended claim? 什么评价标准能够检验预期主张?
5	**Output**	What should the reader receive: estimate, map, interpretation or explanation? 读者最终应得到估计、地图、解释还是机制说明?

Choose the review type only after the decision and evidence product are clear

Core references: JBI Manual for Evidence Synthesis; Cochrane Handbook; RAMESES standards.

Question, Search and Eligibility | 问题、检索与纳入

Review type	Question scope 问题范围	Search and eligibility 检索与纳入
Scoping review 范围综述	Broad field, concepts or methods 宽领域、概念或方法	Broad, iterative search; many designs 广泛且可迭代；可纳入多种设计
Systematic review 系统综述	Focused effect, diagnosis or prognosis 聚焦效果、诊断或预后	Comprehensive search; prespecified criteria 尽量穷尽；预先规定标准
Systematic map 系统证据地图	Distribution of evidence in a large field 大型领域中的证据分布	Broad search; coded categories and study designs 广泛检索；按类别与设计编码

Operational difference 实际差异: a systematic review usually fixes narrower criteria before searching; scoping reviews and maps preserve breadth to describe the field.

Sources: JBI Manual for Evidence Synthesis; Cochrane Handbook for Systematic Reviews of Interventions.

Interpretive Review Workflows | 解释型综述的流程

Review type	Evidence unit 证据单位	How the workflow changes 流程如何改变
Qualitative evidence synthesis **定性证据综合**	Findings, quotations and authors' interpretations 研究发现、引文与作者解释	Sample qualitative studies; code meanings; develop themes or concepts 纳入定性研究；编码意义；发展主题或概念
Realist review **现实主义综述**	Context, mechanism and outcome configurations 情境、机制与结果组合	Search iteratively to test and refine programme theories 围绕项目理论迭代检索、检验与修正
Critical narrative review **批判性叙述综述**	Concepts, assumptions and lines of argument 概念、假设与论证路径	Select and interrogate influential literature; make the interpretive logic explicit 选择并质疑关键文献；明确解释逻辑

Interpretive reviews synthesize meanings, mechanisms or assumptions rather than pooled effects

Sources: JBI Manual for Evidence Synthesis; RAMESES publication standards.

Activity 8: Complete a One-Page Mini-Review Protocol

5 minutes. Draft individually and check with a peer. Use your PhD topic or the running case.

1. Decision and intended output
2. Review type and rationale
3. Review question
4. Eligibility boundaries
5. Information sources

6. Concept blocks and search validation
7. Screening and conflict resolution
8. Extraction and appraisal
9. Synthesis and intended gap statement
10. Responsible AI, disclosure and human verification

Peer check: decision fit, clear boundaries, reproducible search, appropriate appraisal, actionable synthesis and documented AI use

Worked Mini-Protocol and Take-Home Messages

Running-case mini-protocol

Element	Decision
Purpose and review type	Design a multisite validation study. Begin with a systematic map, then a focused prediction-model review.
Question and eligibility	Six-month decline or falls in community-dwelling adults aged 65 years or older. Longitudinal wearable or ambient sensing.
Search	MEDLINE, care or behaviour database, IEEE or ACM, and cross-disciplinary citation coverage.
Selection and appraisal	Piloted rules, independent decisions, recorded reasons, and PROBAST+AI.
Synthesis and output	Claims by validation, utility and equity. Produce a bounded gap and prespecified multisite design.
AI transparency	Record the tool, task, human verification and data-protection decisions.

Take-home messages

- Begin with the research decision
- Fix the boundaries before refining keywords
- Judge credibility before comparing performance
- Define a gap that implies a feasible study

Final output: one page that another researcher can inspect, challenge and reproduce.

The review should support a bounded, testable research decision

A Literature Review Toolchain | 文献综述工具链

Stage 阶段	Find 检索	Manage 管理	Screen 筛选	Extract 提取	Synthesize 综合
Primary task	Identify records	Store PDFs and citations	Apply eligibility rules	Capture comparable data	Form evidence claims
Typical tools	Databases	Zotero / EndNote	Rayyan / Covidence	Excel / review platforms	RevMan / R / qualitative tools

Tools organize work. The protocol defines the question, eligibility rules and human responsibility.
软件组织流程；protocol决定问题、纳排规则与研究者责任。

Choose software after the review design is clear

Search Platforms | 数据库与搜索平台

Source	Best use 适合用途	Important limit 重要限制
PubMed / MEDLINE	Health, medicine and ageing 健康、医学与老龄研究	Limited engineering and computing coverage 工程与计算机覆盖有限
Scopus / Web of Science	Cross-disciplinary search and citation tracking 跨学科检索与引文追踪	Access often depends on institutional subscription 通常需要机构订阅
CINAHL / PsycINFO	Care, nursing, behaviour and experience 照护、护理、行为与体验	Coverage reflects the discipline and indexing policy 覆盖受学科与标引政策影响
IEEE Xplore / ACM DL	Sensors, computing and engineering 传感器、计算与工程	Clinical terminology may be sparse 临床术语可能不足
Google Scholar / CNKI	Supplementary discovery and regional coverage 补充发现与地区性文献	Search transparency and reproducibility vary 检索透明度与可重复性不同

Database choice follows the evidence ecosystem of the question

Reference Management | 文献管理

Tool	Strengths 优势	Good fit 适合情境
Zotero	Free and open source; browser capture; PDF annotation; Word integration 免费开源；网页抓取；PDF批注；Word引用	Default choice for most doctoral researchers 多数博士生的默认选择
EndNote	Mature reference library and Word workflow 成熟的题录库与Word工作流	Teams or institutions already using EndNote 已有EndNote许可与流程的团队
Mendeley Reference Manager	Reference library, PDF reading and citation support 题录库、PDF阅读与引用支持	Researchers already working in its ecosystem 已使用其生态的研究者

A reference manager stores and cites literature. Critical synthesis still requires an explicit analytic method.

Use one reference library as the stable record of the project

Citation Discovery | 引文网络与补充发现

Tool	What it does 功能	Use with care 使用边界
ResearchRabbit	Explores related papers, authors and citation connections 探索相关论文、作者与引文关系	Useful for iterative discovery; preserve the formal database search 适合迭代发现；仍需保留正式数据库检索
Litmaps	Builds citation maps, monitors new papers and links with Zotero 建立引文地图、监测新文献并连接Zotero	A citation map reflects indexed connections, not study credibility 引文连接不代表研究可信度

Best use: start from a verified seed paper, explore backward and forward citations, then record what was added.

Citation tools supplement keyword searching and reveal terminology gaps

Screening and Review Management | 筛选与综述管理

Tool	Core workflow 核心流程	Best fit 适合情境
Rayyan	Deduplication, blind screening, labels, conflicts and exports 去重、盲法筛选、标签、冲突与导出	Accessible starting point for student teams 学生团队的易用起点
Covidence	Screening, full text, extraction, risk of bias and reporting 筛选、全文、提取、偏倚评价与报告	Teams with institutional access and a formal workflow 有机构许可的正式团队
EPPI-Reviewer	Bibliographic management, coding and synthesis 题录管理、编码与综合	Complex evidence synthesis and policy reviews 复杂证据综合与政策综述

The platform records decisions. Reviewers remain responsible for eligibility rules and conflict resolution.

Choose a platform that preserves independent decisions and an audit trail

AI-Assisted Review Tools | AI辅助综述工具

Tool	Useful contribution 有用功能	Required human control 人工控制
ASReview	Active learning prioritizes records during title and abstract screening 主动学习优先呈现可能相关的记录	Define stopping rules and protect recall; humans label records 定义停止规则并保护召回率；人工作出标签
Elicit	Semantic discovery, screening suggestions and structured extraction with supporting text 语义发现、筛选建议和带原文依据的结构化提取	Use traditional searches for publication-grade reviews; verify every extracted claim 正式综述保留传统检索；逐项核对提取结果
Rayyan AI features	Relevance ranking and optional AI-assisted review functions 相关性排序及可选AI辅助功能	Do not delegate the final eligibility decision without a validated protocol 没有验证过的protocol时，不把最终纳排交给AI

AI can prioritize and prefill. Researchers verify sources, decisions and claims

Data Extraction and Qualitative Synthesis | 数据提取与定性综合

Tool	Best use 最适合	Design requirement 方法要求
Excel / Google Sheets	Small or moderate reviews with a transparent extraction matrix 中小型综述的透明提取矩阵	Lock field definitions and pilot the form 固定字段定义并试填表格
Covidence / EPPI-Reviewer	Team extraction, structured forms and audit trails 团队提取、结构化表单与审计记录	Configure fields from the review question 依据综述问题设置字段
NVivo / MAXQDA / ATLAS.ti	Coding qualitative findings, themes and interpretations 编码定性发现、主题与解释	A coding system still needs an explicit analytic method 编码体系仍需明确分析方法
Elicit	Prefills structured fields and links answers to supporting text 预填结构化字段并连接原文依据	Check the source passage and record corrections 核查原文并记录人工修正

Extraction fields should prepare the comparisons needed for synthesis

Quantitative Synthesis and Evidence Mapping | 定量综合与证据地图

Purpose	Tools	What they support
Meta-analysis **元分析**	RevMan	Structured study data, effect estimates, forest plots and reporting 结构化研究数据、效应估计、森林图与报告
Flexible statistical analysis **灵活统计分析**	R: meta / metafor; Stata	Custom models, sensitivity analysis and reproducible code 自定义模型、敏感性分析与可重复代码
Bibliometric networks **文献计量网络**	VOSviewer	Co-authorship, citation, co-citation and keyword networks 作者合作、引文、共被引与关键词网络
Science mapping **科学知识图谱**	Bibliometrix / Biblioshiny	Thematic maps, evolution, collaboration and citation analysis 主题地图、演化、合作与引文分析

A network map describes literature structure. Meta-analysis estimates an effect. Credibility appraisal remains a separate task.

Analysis software follows the synthesis question and the comparability of evidence

Recommended Tool Stacks | 推荐工具组合

Review goal 综述目标	Suggested stack 建议组合	Why this fits 适配原因
Doctoral critical review **博士批判性综述**	Discipline databases + Zotero + ResearchRabbit or Litmaps + Excel	Strong discovery, stable reference library and a transparent synthesis matrix 兼顾发现、管理与透明综合
Scoping review **范围综述**	Multiple databases + Zotero or EndNote + Rayyan or Covidence + Excel	Supports broad retrieval, deduplication, screening and evidence charting 支持广泛检索、去重、筛选与证据制表
Systematic review **系统综述**	Database search + Rayyan, Covidence or EPPI + appraisal tool + RevMan or R when appropriate	Preserves independent decisions, audit trail and planned synthesis 保留独立判断、审计记录与预设综合
Large evidence map **大型证据地图**	Databases + ASReview or EPPI + structured coding + VOSviewer or Biblioshiny when justified	Prioritizes large-scale screening and displays evidence distribution 支持大规模筛选与证据分布展示

The simplest toolchain that preserves transparency is usually the best starting point

Tool Selection and Responsible Use | 工具选择与负责任使用

Question	Decision check 决策检查
Review design	Does the tool support the chosen review type and evidence product? 软件是否支持所选综述类型与证据产品?
Reproducibility	Can the team export queries, decisions, labels and structured data? 能否导出检索式、决定、标签与结构化数据?
Collaboration	Does it support independent reviewers, conflicts and an audit trail? 是否支持独立评审、冲突解决与审计记录?
Data governance	Where are PDFs, unpublished material and personal data stored? PDF、未公开资料与个人数据存放在哪里?
AI verification	Which outputs require source checking and a recorded human decision? 哪些输出必须核对原文并记录人工决定?
Exit strategy	Can the project export data in a reusable format if access ends? 权限结束后能否以可复用格式导出项目?

Software should make the review easier to inspect, challenge and reproduce



AIHA8121: Research Methodology & Ethics

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